

General Clustering Features:

customer_id: Unique identifier for each customer record.

kyc_type: Customer classification type (e.g., Individual or Business) based on KYC records.

province: Registered province of the customer.

city: Registered city of the customer.

account_age: Number of days since the account was opened.

birth_business_age: Age of the individual (if personal account) or business age (if corporate account), in years.

total_transaction_count: Total number of transactions recorded for the customer.

transaction_count_ABM: Number of transactions conducted via ABM.

transaction_count_Card: Number of card-based transactions.

transaction_count_Cheque: Number of cheque transactions.

transaction_count_EFT: Number of Electronic Funds Transfer transactions.

transaction_count_EMT: Number of Email Money Transfer transactions.

debit_transaction_amount: Total value in cents of debit transactions.

credit_transaction_amount: Total value in cents of credit transactions.

average_debit_transaction_amount: Average amount in cents per debit transaction.

average_credit_transaction_amount: Average amount in cents per credit transaction.

unique_transaction_cities: Number of distinct cities where transactions occurred.

unique_transaction_provinces: Number of distinct provinces where transactions occurred.

multi_country_transactions: Indicator or count reflecting whether transactions occurred across multiple countries.

card_ecommerce_count: Number of card transactions conducted through e-commerce channels.

card_non_ecommerce_count: Number of card transactions conducted in (non-e-commerce).

card_ecommerce_to_nonecommerce_ratio: Ratio of e-commerce card transactions to non-e-commerce card transactions.

card_mcc_diversity_count: Number of distinct Merchant Category Codes (MCCs) used in card transactions.

card_top_mcc_weight: Proportion of card transactions concentrated in the most frequent MCC.

card_top_mcc: Most frequently used Merchant Category Code for the customer.

Averagetime_between_transaction_day: Average number of days between transactions.

std_time_between_transactions: Standard deviation of time intervals between transactions.

median_time_between_transactions: Median number of days between transactions.

pct_transactions_top10_busiest_days: Percentage of transactions concentrated within the customer's top 10 busiest transaction days.

economic_role: Engineered categorical variable representing economic classification (e.g., industry or occupation).

credit_share: Proportion of total transaction amount that comes from credit transactions.

ABM_cash_weight: Proportion of total transactions conducted via cash ABM relative to overall ABM transaction activity.

Datasets(csv):

clustering_cleaned: Dataset derived from master_customer_dataframe for use in all clusterings.

Version A folder:

Clustering_Business_Plots: Folder containing visualization outputs for business-only clustering results, including feature heatmaps, distribution plots, and cluster comparison charts.

Clustering_Individual_Plots: Folder containing visualization outputs for individual-only clustering results, including behavioral pattern plots and feature heatmaps by cluster.

Cluster Profile: Document summarizing the characteristics of each cluster (both business and individual), including high-level behavioral segmentation descriptions.

clustering_business_only.ipynb: Jupyter Notebook implementing the clustering workflow specifically for business customers, including preprocessing, FAMD transformation, and model fitting.

clustering_individual_only.ipynb: Jupyter Notebook implementing the clustering workflow specifically for individual customers, including preprocessing, FAMD transformation, and model fitting.

Kmeans_Seperate.csv: Output dataset containing final cluster labels generated after clustering the two datasets separated by KYC type.

VersionA_Business_AfterFAMD.csv: Business-only dataset after FAMD transformation, containing component scores used for clustering.

VersionA_Business_BeforeFAMD.csv: Business-only engineered feature dataset prior to dimensionality reduction.

VersionA_Individual_AfterFAMD.csv: Individual-only dataset after FAMD transformation, containing component scores used for clustering.

VersionA_Individual_BeforeFAMD.csv: Individual-only engineered feature dataset prior to dimensionality reduction.

Version B folder:

Clustering_VersionB_Output: Folder containing output artifacts from the Version B clustering pipeline (e.g., cluster labels, evaluation metrics, plots, or processed datasets).

Cluster Profile: Document file summarizing cluster characteristics, including behavioral patterns for each cluster.

Kmeans_Clustering.ipynb: Jupyter Notebook implementing the K-Means clustering workflow, including preprocessing, FAMD transformation, model fitting, and evaluation steps.

Kmeans_two_round.csv: Dataset containing final cluster labels after the two-round clustering process (initial global clustering + reclustering of business dominant cluster).

VersionB_afterFAMD.csv: Processed dataset after FAMD transformation, containing numerical component scores used as input for clustering.

VersionB_BeforeFAMD.csv: Original engineered feature dataset prior to dimensionality reduction, containing raw numerical and categorical variables prepared for FAMD.

